

Effect of Ambulatory Conditions and Virtual Locomotion Techniques on Distance Estimation and Motion Sickness of a Navigated VR Environment

Aidan G. Morris, Michael C. Vail, Gabriel S. Hanna, and Anurag Rimzhim*

College of the Holy Cross

ABSTRACT

We present preliminary results from a 2 X 2 between-subjects experiment. Our two IVs were ambulatory-restrictive (i.e., without locomotion) postural conditions (sitting vs. standing), and virtual navigation techniques (steering vs. teleporting). Participants navigated a complex virtual environment comprising outdoor and indoor environments for 10 minutes. We found that teleporting may result in less online distance estimation error than steering. Motion sickness was lower while teleporting than steering and when sitting than standing. Teleporting also resulted in better system usability than steering. We discuss the results' implications for VR usability.

Keywords: Virtual reality, HMD, locomotion, distance estimation, motion sickness.

1 INTRODUCTION

Ambulatory and podokinetic information influence spatial learning in virtual environments [1]. Navigating a virtual space while sitting results in similar or worse performance when using an HMD VR (head-mounted display virtual reality) than a desktop VR on spatial memory tasks [2]. This result was explained by higher motion sickness and the absence of locomotion-based cues in the sitting HMD VR versus sitting desktop VR condition [2]. The current study tests these findings further using a 2 X 2 between-subjects design. We used one of the same VR settings (i.e., HMD VR) as in [2] but different ambulatory-restrictive conditions. We chose ambulatory-restrictive (i.e., without locomotion) postural conditions (i.e., sitting, SI, vs. standing, ST) as the first IV (termed as *ambulatory condition*) because we believe these are the two most commonly used postures that people adopt while using VR. Sitting and standing are similar due to the absence of locomotion but differ in the vestibular information that can help spatial navigation.

Locomotion techniques can affect virtual navigation, with teleporting resulting in better spatial learning and lower motion sickness than steering [3] [4]; presence and system-usability generally are not affected by this factor [3] [4]. The second IV in our study was the *virtual locomotion technique* with two levels: steering (STR) versus teleporting (TLP). We chose the above two IVs because they commonly interact in real life for VR users and have not been studied using a factorial design. For dependent variables, we measured spatial learning using distance estimation, both while the participants were in the virtual environment (online) and after virtual navigation (offline).

We also measured presence, motion sickness, and system usability.

We hypothesized that distance estimation would be better in standing versus sitting and in teleporting versus steering conditions. We also hypothesized that motion sickness would be lower and system usability would be higher in standing versus sitting conditions (due to better idiothetic information) and in teleporting versus steering conditions (due to less conflicting spatial information).

2 METHOD

Seventy-six participants, most undergraduate students at the College of the Holy Cross, MA, voluntarily agreed to participate in the study for course credit or monetary compensation of a \$15 gift card. Participants were equally and randomly distributed across the four experimental conditions.

We used Oculus Quest 2 as the HMD VR for virtual navigation. The virtual environment (see Figures 1, 2, & 3) included both outdoor and indoor environments. It was modified and rendered using Unity3D at 60 frames per second. It consisted of two major landmasses separated by a bridge. It contained many buildings and objects but had no living beings.

The procedure involved giving participants some practice in navigating the virtual environment after familiarizing themselves with the VR headset and hand controllers. The main trials started with participants rating the familiarity of pictures of five critical items (mixed with five control items) that they would later see in the virtual environment. We did this to ensure no objects were novel to the participants. They navigated the virtual environment for a maximum of 10 minutes, starting from the same starting point. They were instructed that their task was to navigate the virtual environment well enough so that they could later answer questions about its spatial layout and the objects they encountered. They were also told to return to the starting point by or before 10 minutes of navigation. Before starting the navigation, they estimated three distances between objects (online distance estimation). Immediately after navigation, they filled out surveys on motion sickness, presence, and system usability in that order. They then engaged in offline distance estimation of the distances they recalled earlier. They were instructed to rethink the distances they estimated earlier and if they would change their previous answers based on the virtual navigation. They then completed an object recognition task and filled out a demographic form. The full procedure took around an hour.

*Corresponding author: arimzhim@holycross.edu



Figure 1: Spatial Layout of the Virtual Environment.



Figure 2: Example of an Outdoor Virtual Environment.



Figure 3: Example of an Indoor Virtual Environment.

3 RESULTS

We conducted between-subject 2 X 2 ANOVA for each dependent variable separately. For *online distance estimation error*, neither ambulatory condition ($p = .960$), locomotion condition ($p = .099$), nor interaction was significant ($p = .069$). However, the numerical trend showed that the difference in distance estimation error between TLP (79 feet) and STR (110 feet) was higher (31 feet) when standing than when sitting (2 feet; TLP: 96 feet; STR: 94 feet). See Figure 4.

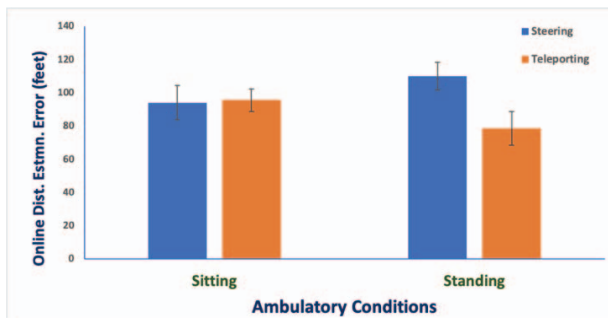


Figure 4: Online Distance Estimation Error.

For *motion sickness*, both ambulatory condition ($F(1, 72) = 4.696$, $p = .034$, $\eta_p^2 = .061$) and locomotion condition ($F(1, 72) = 26.340$, $p < .001$, $\eta_p^2 = .268$) had main effects, but there was no interaction ($p = .110$). See Figure 5.

For *system usability*, only the locomotion condition had a main effect ($F(1, 72) = 4.869$, $p = .031$, $\eta_p^2 = .063$); there was no interaction ($p = .359$). See Figure 6.

For *presence*, neither the ambulatory condition ($p = .924$), locomotion condition ($p = .677$), nor the interaction was significant ($p = .878$).

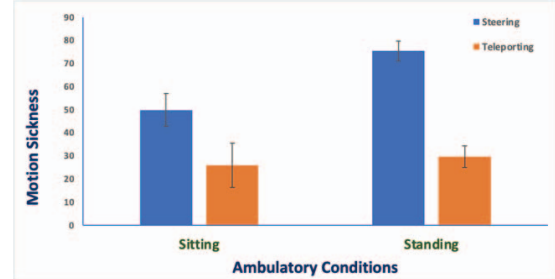


Figure 5: Motion Sickness.

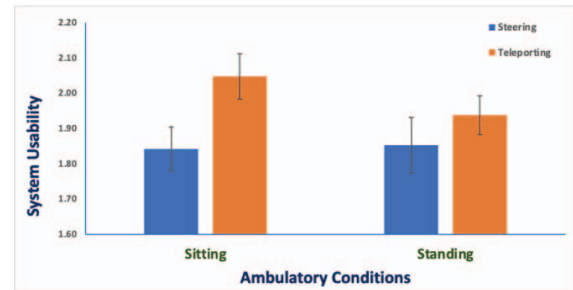


Figure 6: System Usability

4 DISCUSSION

Our preliminary results show that teleporting is better than steering as a virtual locomotion technique because it results in lower motion sickness and higher system usability. It may also reduce errors when estimating distance in a virtual environment using HMD VR. These results replicate previous findings [2] [3][4] in a novel, complex virtual environment that has both indoor and outdoor environments. We also found that using HMD while sitting results in less motion sickness than standing.

The above results provide evidence that HMD VR's application and usability can be enhanced by using teleporting versus steering locomotion while sitting rather than standing.

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